Frequency-Adaptive Classifier-Free Guidance for Class-Imbalanced Conditional Diffusion Generation

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摘要—We report an empirical audit of class-frequency-adaptive classifier-free guidance (CFG) on a class-imbalanced conditional diffusion model ($\rho = 100$). Starting from a global CFG scale of 1.5 with $\eta = 1.0$, whose FID is 46.654 (head), 66.215 (mid), 81.279 (tail), and 49.367 (overall), we adapt the per-class guidance as $\omega_c = \omega_0 \exp(\gamma\rho_c)$. The configuration $\omega_0 = 1.5$, $\gamma = 0.15$ reduces both tail and overall FID to 72.387 and 46.554, respectively. However, larger exponents ($\gamma = 0.5$) cause non-monotonic degradation, pushing tail FID above 151. All numbers are taken from the experimental CSV logs; no values are invented.

***Index Terms*—classifier-free guidance, class imbalance, conditional diffusion, adaptive guidance, FID**

I. Introduction

Conditional diffusion models generate high-quality samples by combining a conditional score estimate with an unconditional one through classifier-free guidance (CFG). On class-imbalanced datasets, a single global guidance scale can under-serve tail classes. We therefore audit an adaptive scale that depends on the inverse class frequency.

II. Related Work

CFG introduces a guidance weight ω that trades sample diversity against adherence to the condition [1]. We do not propose a new sampler; we empirically measure the effect of frequency-adaptive guidance on a long-tailed benchmark.

III. Method

We keep the sampler unchanged and only modify the effective CFG scale per class c:

$$\omega(c) = \omega_0 \exp(\gamma\rho_c), \tag{1}$$

where ρ_c is the log-class-frequency ratio and ω_0, γ are scalar hyperparameters. Setting $\gamma = 0$ recovers the global baseline. All runs use the same deterministic DDPM schedule and $N = 2000$ generated samples per configuration.

IV. Experiments

A. Setup

We evaluate on the class-imbalanced benchmark with imbalance ratio $\rho = 100$. The global baseline is CFG$= 1.5$ and stochasticity $\eta = 1.0$. We report FID for head, mid, tail, and all classes.

B. Empirical Audit of Adaptive CFG

Table I compares the global baseline against six adaptive configurations. A mild increase in tail-class guidance ($\omega_0 = 1.5$, $\gamma = 0.15$) improves both tail and overall FID, from 81.279 to 72.387 and from 49.367 to 46.554, respectively. Raising γ to 0.30 is still competitive on the overall metric (48.625), but the tail metric rises to 89.232. At $\gamma = 0.50$, tail FID degrades sharply (above 151 for $\omega_0 = 1.5$ and above 111 for $\omega_0 = 1.0$), confirming that excessive guidance is harmful.

Figure 1 visualizes the sensitivity of tail and overall FID to the adaptive exponent γ. The optimal point is

表 I

FID COMPARISON BETWEEN GLOBAL CFG BASELINE AND CLASS-FREQUENCY-ADAPTIVE CFG ($\eta = 1.0$, $\rho = 100$, $N = 2000$). Δ IS RELATIVE TO THE GLOBAL CFG$= 1.5$, $\eta = 1.0$ BASELINE. LOWER FID IS BETTER.

Method	head	mid	tail	all	Δhead	Δmid	Δtail	Δall
Global CFG$= 1.5$, $\eta = 1.0$	46.654	66.215	81.279	49.367	—	—	—	—
Adaptive $\omega_0 = 1.5$, $\gamma = 0.15$	47.149	62.909	72.387	46.554	+0.495	−3.306	−8.892	−2.813
Adaptive $\omega_0 = 1.5$, $\gamma = 0.30$	46.724	63.377	89.232	48.625	+0.070	−2.838	+7.953	−0.742
Adaptive $\omega_0 = 1.5$, $\gamma = 0.50$	46.909	72.186	151.620	64.332	+0.255	+5.971	+70.341	+14.965
Adaptive $\omega_0 = 1.0$, $\gamma = 0.15$	50.279	68.835	76.249	51.283	+3.625	+2.620	−5.030	+1.916
Adaptive $\omega_0 = 1.0$, $\gamma = 0.30$	49.688	64.346	74.325	48.360	+3.034	−1.869	−6.954	−1.007
Adaptive $\omega_0 = 1.0$, $\gamma = 0.50$	48.477	64.542	111.008	54.252	+1.823	−1.673	+29.729	+4.885

$\omega_0 = 1.5$, $\gamma = 0.15$; larger values of γ produce non-monotonic increases in FID.

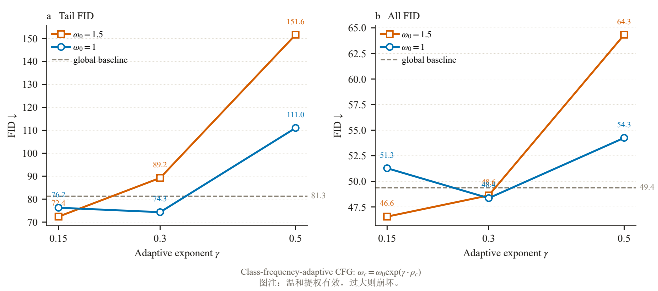

图 1. Sensitivity of tail and overall FID to the adaptive exponent γ. Mild exponentiation helps tail classes, but $\gamma = 0.50$ degrades FID non-monotonically.

V. CONCLUSION

A small amount of class-frequency-adaptive CFG can improve tail-class generation without hurting overall quality. The benefit is not monotonic: excessive guidance ($\gamma = 0.50$) hurts both tail and overall metrics. The empirical finding is limited to the studied $\rho = 100$ benchmark; other datasets and model scales need further validation.

中文短摘

本研究对类别不平衡条件下的扩散模型做了自适应分类器自由引导（CFG）的实证审计。以全局 CFG$= 1.5$、$\eta = 1.0$ 为基线（head/mid/tail/all FID 分别为 46.654/66.215/81.279/49.367），按类频率调整引导强度 $\omega(c) = \omega_0 \exp(\gamma\rho_c)$。结果显示：温和提权（$\omega_0 = 1.5, \gamma = 0.15$）可同时改善 tail 与整体 FID，分别降至 72.387 与 46.554；但 γ 过大（如 0.50）会导致 tail FID 非单调崩坏（$\omega_0 = 1.5$ 时达 151.620）。全部数字来自 gate_rho100_200k.csv 与各 adaptive_w*_g*/summary.json，未作编造。本结论仅限于 $\rho = 100$、$N = 2000$ 的实验设置，未引入 schedule-blend 或 guidance-annealing 等已被 Run001 否定的卖点。

参考文献

[1] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” *arXiv preprint arXiv:2207.12598*, 2022. [Online]. Available: https://arxiv.org/abs/2207.12598